

Unit 1.2 Practice Packet

Program Function & Purpose



Practice Makes Prepared

Parts 1–2 are vocabulary that shows up directly on the AP exam — know these words cold. Parts 3–8 apply them the way the exam does. Use your guided notes and vocab sheet if you get stuck.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|--|--|
| _____ 1. Program | A. Data sent from a program to a device (text, visuals, audio, or tactile) |
| _____ 2. Code segment | B. A collection of program statements that is part of a program |
| _____ 3. Behavior | C. Why a program exists: the problem it solves or the creative interest it pursues |
| _____ 4. Event | D. A device that measures the physical world (temperature, light, motion) and sends it as input |
| _____ 5. Event-driven programming | E. How a program functions during execution; often described by how a user interacts with it |
| _____ 6. Input | F. Program statements run when triggered by an event, not in a set order |
| _____ 7. Output | G. The screens, buttons, and controls a user interacts with |
| _____ 8. Purpose | H. A collection of program statements that performs a specific task when run (also called software) |
| _____ 9. Function | I. Output the user can feel, such as a vibration |
| _____ 10. Sensor | J. Data sent to a computer for processing |
| _____ 11. User interface | K. What a program does when it runs |
| _____ 12. Tactile output | L. An action that supplies input data to a program, such as a click or key press |

Part 2 · Fill It In

Use each word from the word bank once.

input

output

event

purpose

function

sensor

code segment

behavior

1. A motion detector that turns on a porch light is an example of a _____.
2. The four lines inside a procedure that calculate a tip are a _____ of the larger app.
3. "This app helps students find study partners" describes the app's _____.
4. Pressing the jump button in a game is an _____ that triggers code to run.
5. The alarm sound a timer app plays when time is up is an _____.
6. Explaining what happens when a user taps, swipes, or types describes the program's _____.
7. "The app sorts your tasks by due date" describes a _____.
8. Typing your name into a sign-up form provides _____ to the program.

Name: _____

Class: _____

Date: _____

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Inputs, Outputs & Events

Part 3 · Input, Output, or Event?

Check the best label for each. An **event** is a user action or trigger that makes code run (it is also a kind of input).

Situation	Input	Output	Event
1. A student taps “Add to Cart”			
2. A fitness band buzzes on your wrist			
3. A thermometer reading sent to a smart thermostat			
4. A map app shows the fastest route on screen			
5. A voice assistant hears “What’s the weather?”			
6. A game controller rumbles when you crash			
7. A timer reaching 0:00 starts a buzzer			
8. A security camera’s video feed sent to an app			
9. A printer printing a report			
10. A user presses the spacebar to jump			

Part 4 · Purpose or Function?

Write **P** or **F**.

Statement	P/F
1. A budgeting app helps teens save money for things they want.	
2. The app adds up all purchases each week.	
3. A study app sends a reminder at 6:00 PM.	
4. A map app helps wheelchair users find accessible entrances.	
5. A game lets players create and share their own levels.	
6. A language app makes learning Spanish fun for beginners.	

Part 5 · Fix It!

A student wrote: “*The purpose of my app is to show a map with pins on it.*”

(a) Why is this really a **function**, not a purpose?

(b) Rewrite it as a true purpose statement.

Name: _____

Class: _____

Date: _____

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Analyze Programs Like the Exam

Part 6A • Bus Tracker

A bus-tracker app uses GPS on each bus to show its location on a map. When a student taps their stop, the app shows how many minutes until the bus arrives and vibrates the phone when the bus is 2 minutes away.

	Your answer
Purpose	
Input(s)	
Output(s)	
Which input is an event?	

Part 6B • Plant Helper

A plant-care app uses a soil-moisture sensor. Every hour it checks the soil. If the soil is dry, it plays a chime and displays “Water me!” Users can tap “Watered” to reset the reminder.

	Your answer
Purpose	
Input(s)	
Output(s)	
Which input is an event?	

Part 7 • Events in Code

height starts at 0.

```
ON keyPress("space")
{
  height ← height + 10
  DISPLAY(height)
}
ON keyPress("r")
{
  height ← 0
  DISPLAY(height)
}
```

1. Name the two events in this program:
_____ and _____.
2. The user presses **space, space, r, space**.
What is displayed? _____
3. What runs if the user presses **q**?

4. Is this program sequential or event-driven?



Think:

Remember the sentence frame: “When the user _____, the program _____.” The first blank is the input/event, the second is the output.

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Exam Practice

Part 8 • Exam Practice

★ Warm-up

1. Which is an example of **tactile** output?
- A. A message displayed on a screen
 - B. A beep from a speaker
 - C. A phone vibrating when a text arrives
 - D. A photo taken by a camera

★★ On Level

2. A parking-garage program uses sensors to count cars entering and leaving, then shows the number of open spots on a sign. Which is an **output**?
- A. The number of open spots shown on the sign
 - B. The sensor detecting a car
 - C. A car entering the garage
 - D. The count stored in memory

★★ On Level

3. A student describes their app: "When the user taps *Shuffle*, the program displays a random quote." Which part is the **event**?
- A. Displaying a quote
 - B. The random number
 - C. The list of quotes
 - D. Tapping Shuffle

★★★ Exam Hard

4. Which statement best describes the **purpose** (not a function) of a recycling app?
- A. It scans barcodes using the camera
 - B. It helps residents reduce landfill waste by showing which items can be recycled
 - C. It shows a green check mark for recyclable items
 - D. It saves a history of scanned items

★★★ Exam Hard

5. Consider this code segment:

```
ON timerEnds()
{
  DISPLAY("Time's up!")
  playSound("bell")
}
```

Which TWO statements are true? (*Select two.*)

- A. The timer ending is an event.
- B. The message and the bell sound are both outputs.
- C. The code runs as soon as the program starts.
- D. playSound is an input.

Create Task Connection

Describe an app you would build for your school. In 3–4 complete sentences, state its **purpose**, describe one **function**, identify one **input** (and say whether it is an event), and one **output**.
